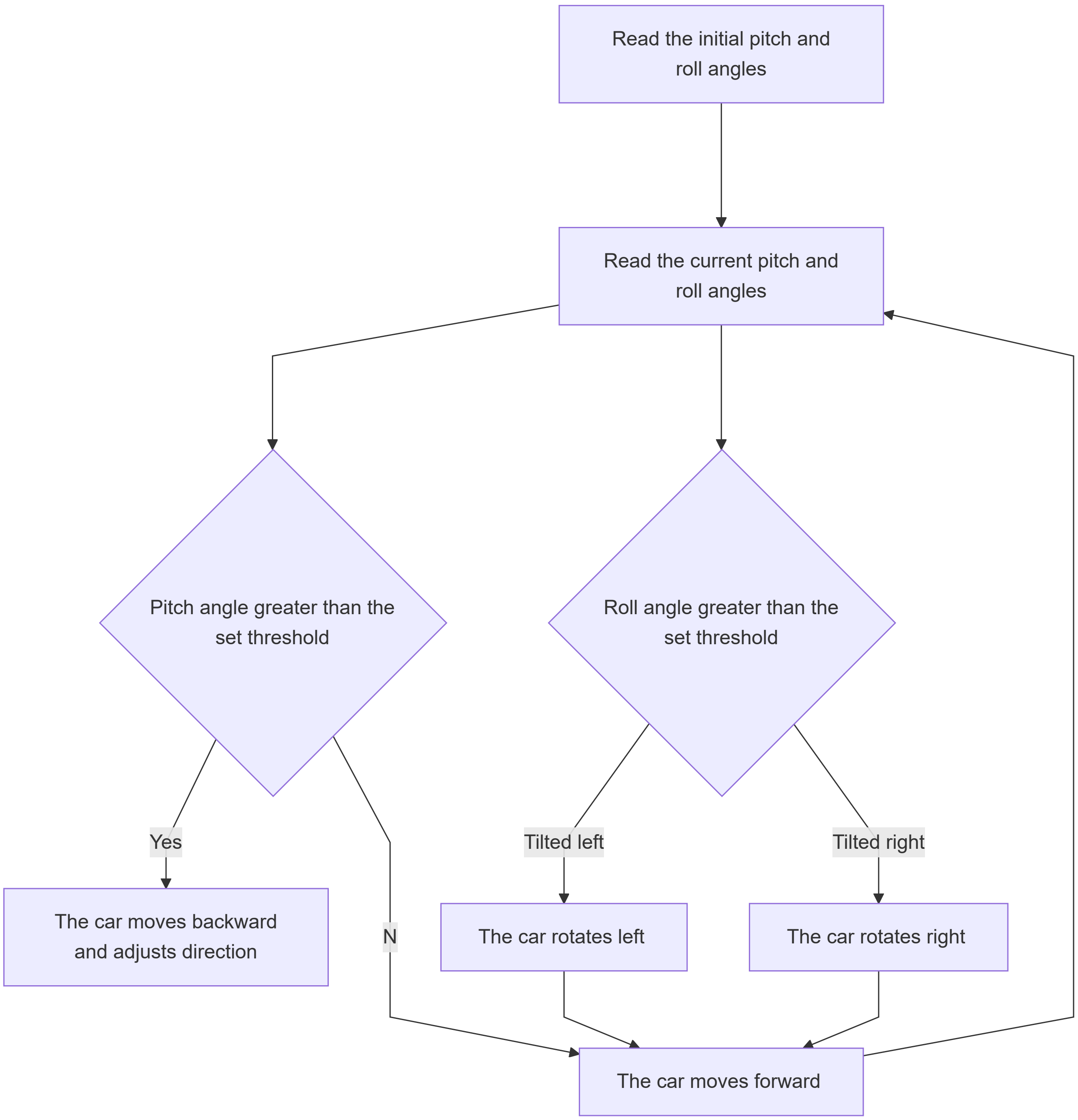
Flatland Exploration Vehicle

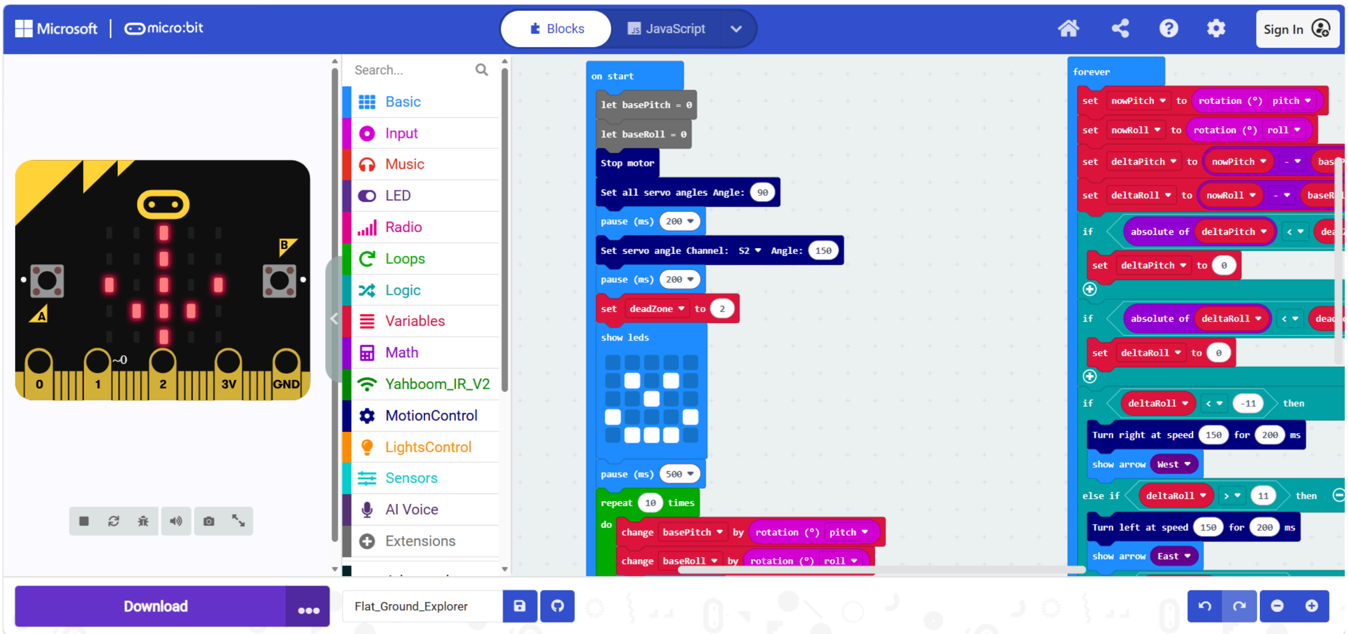
1. Experiment Purpose

In this section, we will use the Micro:bit angle sensor together with motor control to achieve the "flat ground explorer" effect.

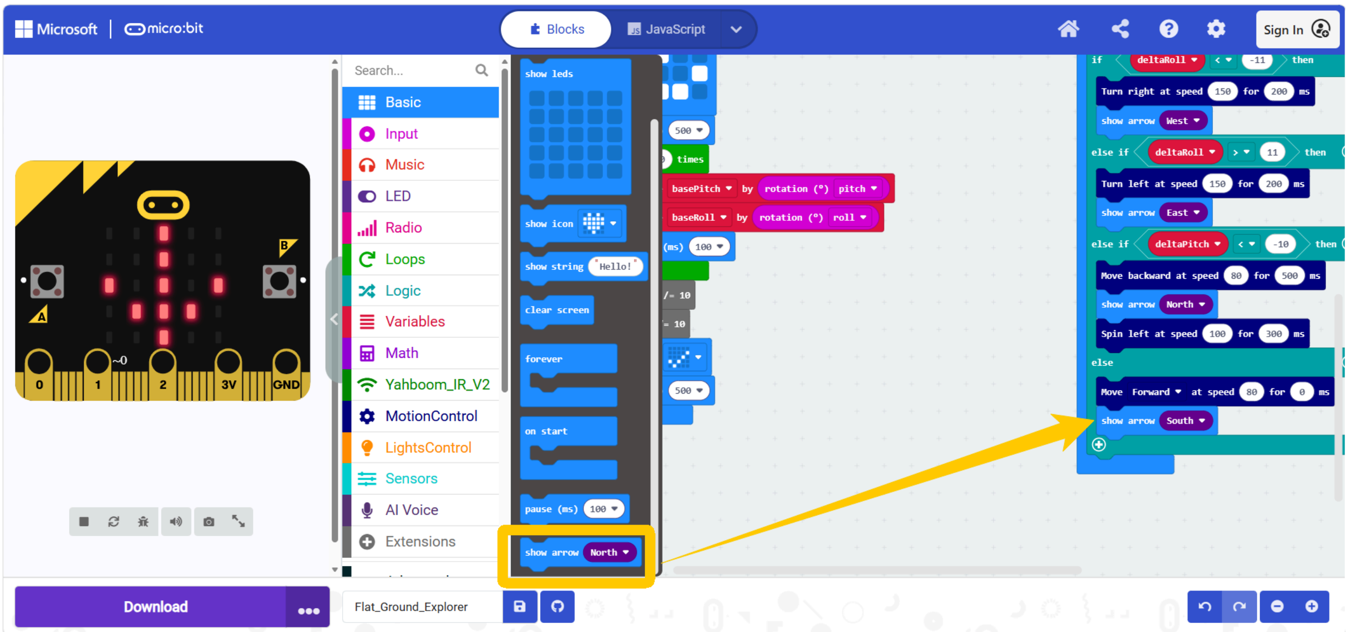
2. Implementation Principle



The program is as follows.



The show arrow block is located in the `Basic` category.



The positions of other blocks have already been introduced in previous tutorials, so we will not go into too much detail here.

4. Experiment Steps

1. Before flashing, turn off the car switch (switch it to OFF), and connect the Micro USB cable to the Micro:bit. Note: not the Charging interface on the expansion board.
2. Flash `12.Flatland_Exploration_Vehicle.hex` to the Micro:bit. For the flashing method, refer to section "4. Flashing Programs" in "2. Introduction to Graphic Programming"
3. Turn on the car switch (switch it to ON).

5. Experiment Phenomenon

The car will detect the current angle information of the car. Lift up the front of the car; when the detected lift angle reaches the threshold, the car moves backward, turns left to adjust its heading, and then moves forward to look for flat ground. When tilted to the left, the car turns left to adjust its heading (when tilted to the right, the car turns right to adjust its heading), and then moves forward, just like an explorer looking for flat ground. The angle threshold can be adjusted according to the actual situation.

This case depends on the terrain, and the threshold needs to be adjusted multiple times. It is recommended to use a thin book to simulate a slope road condition for the experience.